SKITII

Clinical Trials Platform

Product Story & Technical Specification

VERSION	1.0
DATE	July 2027
CUSTOMER	Yenepoya Hospital
SCOPE	React Native Tablet App + Trial Backend + Trial Admin Panel + Existing Auth Backend

Confidential -- Internal Use Only

1. Overview

Skitii Trials is a clinical research version of the Skitii product, built for running music-therapy trials at hospitals. The first customer is Yenepoya Hospital.

- The trial app is almost identical to the Skitii product in session flow (Pre Therapy -> Baseline -> Live Session -> Post Therapy).
- The key difference is that the trial collects extra patient clinical data (cancer staging, ECOG, weekly psychological assessments) before and during the study.
- After the trial concludes, the data and learnings will feed back into the product.

2. Architecture

2.1 Systems involved

System	Role
Existing Skitii Auth Backend	Email OTP login, JWT tokens, Institution identity
New Trial Backend	All trial domain data (patients, sessions, weekly reports,
New Trial DB (MongoDB)	Separate from product DB
New Trial Admin Panel	Admin interface for managing trial data, patients, and reports
Existing Skitii Admin Frontend	Admin adds controlPatientMusic flag to music assets
Trial React Native App	Tablet-only (landscape), Yenepoya institute
Polar H10	BLE heart rate monitor used during sessions and weekly HRV

2.2 Auth flow

1. Trial app calls Existing Auth Backend -> email OTP -> JWT.
2. JWT is passed to Trial Backend for all domain API calls.
3. Trial Backend validates JWT and extracts institutionId to scope all data.
4. Yenepoya is registered as an Institution with product: "skitii-trial" in the Auth DB.

2.3 Multi-device

- One institute (Yenepoya) logs in on multiple tablets.
- PIN is device-local (stored in device Keychain/Keystore only).
- Each tablet creates its own PIN after OTP login.
- Trial Backend must support concurrent writes from multiple tablets under the same institutionId.
- Device restrictions follow the same mechanism as the Skitii product -- no separate implementation for the Trial app.

3. Trial Structure

3.1 Trial duration

- Each patient is enrolled for either 3 weeks or 6 weeks (set at enrollment, field: expectedDuration).
- Completion date = enrollmentDate + 21 days (3-week) or enrollmentDate + 42 days (6-week).
- After completion date the patient cannot start any new sessions.

- There is no restriction on the number of sessions within the trial period.

3.2 Weekly form calendar

- Study weeks run Monday to Sunday (calendar week, not rolling 7 days).
- Week 0 form: submitted before the patient's first session starts. Mandatory.
- Week 1 onwards: submitted in the calendar week following the study week being reported.

Study period	Session days	Weekly form window
Week 0	Before first session	Before first session starts (mandatory)
Week 1	17 Jul (Fri) - 23 Jul (Thu) 2027	24 Jul (Fri) - 2 Aug (Sun) 2027
Week 2	24 Jul (Fri) - 30 Jul (Thu) 2027	3 Aug (Mon) - 9 Aug (Sun) 2027
Week 3	31 Jul (Fri) - 6 Aug (Thu) 2027	10 Aug (Mon) - 16 Aug (Sun) 2027

How the form window is calculated: Each session week runs for 7 days from the enrollment date. The form window for that week opens on the Monday after the session week ends and runs until the following Sunday. The final week's form window extends beyond the completion date -- this is intentional; the patient can still submit after sessions are complete.

- Completion = 10 Aug 2027 (July 21 is day 1, August 10 is day 21).
- Session weeks: 21 Jul (Tue) - 27 Jul (Mon) / 28 Jul (Tue) - 3 Aug (Mon) / 4 Aug (Tue) - 10 Aug (Mon) 2027.
- Form windows: 28 Jul (Tue) - 9 Aug (Sun) / 10 Aug (Mon) - 16 Aug (Sun) / 17 Aug (Mon) - 23 Aug (Sun) 2027.
- Patient can submit the final week form even after the completion date.

3.3 Email reminder

- A cron job on the Trial Backend sends an email to the institute's login email on Thursday, Friday, and Saturday of each weekly form window (reminding before the Sunday deadline).
- Reminder includes which patients have a pending weekly form for that week.

4. Session Phases

4.1 Three daily phases

Every patient must complete 3 sessions per day, one per phase:

Phase	Window	Day attribution
Phase 1	4:00 AM -> 12:00 PM	Same calendar day
Phase 2	12:00 PM -> 8:00 PM	Same calendar day
Phase 3	8:00 PM -> 3:59 AM (next day)	The day Phase 3 started (e.g. session at 2AM on July 18 = Phase 3 of July 17)

A clinical "day" for this trial runs from 4:00 AM to 3:59 AM (next calendar day).

4.2 Phase ordering rules

1. Phases must be completed in strict order: Phase 1 -> Phase 2 -> Phase 3.
2. Exception: If Phase 1 is missed (window closed), Phase 2 is still allowed.
3. A session that was missed in its phase cannot be started in a later phase.
4. Backend enforces all phase ordering and time-window validation.

4.3 Missed session

- A session is missed if:
  - Its phase window has closed, AND
  - No completed Post Therapy form was submitted for that phase.
- Missed sessions count toward compliance statistics.

4.4 Session auto-complete

- Auto-complete works the same as the Skitii product (timer-based).
- Frontend sends to backend whether the session ended by user_stopped or auto_completed.

5. Session Flow

Each session follows a linear 3-step flow with a stepper (Pre Therapy -> Therapy Session -> Post Therapy):

```
Dashboard (tap Action chip)
    ?
Pre Therapy Form
    ?
Baseline Measurement (live HR from Polar H10)
    ?
Live Therapy Session (BLE + music)
    ?
Post Therapy Form
    ?
Dashboard (Action chip -> Done / Restart / Resume)
```

5.1 Pre Therapy Form

Collected before every session:

Vital Signs

- Blood Pressure Systolic* (mmHg)
- Blood Pressure Diastolic* (mmHg)
- Baseline Heart Rate (auto-filled from live Polar reading, read-only)

Patient Assessment

- Current Pain Level* (slider 0-10)
- Energy Level* (Good / Fair / Poor)
- Comorbidities Alert (optional text)
- Medication Changes since last session (optional text)

Session Setup

- Headphone Type* (Over-Ear / In-Ear)
- Posture* (Seated / Supine)
- Quiet Room Verified* (Yes / No)
- Session Environment Observations (optional text)

CTA: Start Session (disabled until required fields valid + Polar HR available)

5.2 Baseline Measurement

- Countdown timer (duration TBD -- same as product).

- Live HR + SD of HR from Polar H10 displayed.
- Heart rate waveform shown.
- "Please remain still and relaxed" instruction.
- Auto-advances to Live Session when timer completes.
- Stores: startedAt, endedAt, durationSec, readings[{hr, timestamp}], meanHr, sdHr.

5.3 Live Therapy Session

For Adaptive patients -- same as Skitii product:

- Music selected by HRV/HR-tier algorithm.
- Music ranking updated per phase based on RMSSD change.
- Phases cycle with HRV rolling window.

For Control patients -- different:

- Music pool = tracks flagged as controlPatientMusic: true in Skitii Admin.
- On session start: one random track from the pool plays continuously until session end.
- If patient changes music: another random track from the pool plays continuously until session end.
- Patient can change music anytime (unlimited changes).
- Ranking is NOT updated for control patients.
- Session readings store multiple tracks per phase (one entry per music change):

[{ trackId, startedAt, endedAt }, ...] per phase.

- Device HR (bpm), HRV RMSSD, SD HR, AVG RR (150s), PNN50, SDNN, Artifact %
- Phase number, HRV tier, time remaining, session timer, REC indicator
- Patient name, LIVE badge, Polar battery

5.4 Post Therapy Form

Collected after every session (or after Stop):

Vital Signs

- Blood Pressure Systolic / Diastolic (mmHg)
- Baseline Heart Rate (from pre-session/baseline, read-only display)

Patient Assessment

- Current Pain Level* (slider 0-10)
- Energy Level* (Good / Fair / Poor)
- Session Completion Status* (dropdown -- enum TBD)

Observations

- Adverse Events* (toggle No/Yes + notes text if Yes)
- Technical Issues* (toggle No/Yes + notes text if Yes)

Data Summary

- RMSSD Change* (auto-filled from session, not manually editable)
- Average RMSSD* (auto-filled from session, not manually editable)
- Session Notes (optional text)

CTA: Complete Session (disabled until required fields valid)

6. Weekly Assessment Flow (5 Steps)

Triggered from Dashboard Weekly -> Start.

```
Dashboard (tap Weekly Start)
    ?
Step 1 -- Patient Info + Vitals
    ?
Step 2 -- VAS (Visual Analog Scale)
    ?
Step 3 -- NCCN Distress Thermometer + Problem Checklist
    ?
Step 4 -- HADS (Hospital Anxiety & Depression Scale)
    ? (Submit all 4 steps together in one API call)
Step 5 -- 10-Minute Resting HRV (Polar H10, submitted separately)
    ?
Dashboard (Weekly chip -> Done)
```

6.1 Step 1 -- Patient Info + Vitals

- Patient Name (read-only, from Dashboard selection)
- Study Week* (dropdown: Week 0, Week 1, ..., Week N)
- Blood Pressure Systolic / Diastolic (mmHg)
- Respiratory Rate* (breaths/min)
- Temperature* (degC)
- Weight* (Kg)
- ECOG Performance Status* (dropdown)

6.2 Step 2 -- VAS (Visual Analog Scale)

Four 0-10 sliders (must be explicitly touched -- default 5 does not count as answered):

1. How stressed do you feel right now? (No stress -> Worst possible stress)
2. How well did you sleep in the past 24 hours? (Worst -> Best)
3. How would you rate your overall well-being? (Worst -> Best)
4. How anxious do you feel about your cancer treatment? (No anxiety -> Extreme anxiety)

6.3 Step 3 -- NCCN Distress Thermometer + Problem Checklist

- Distress level (0-10 slider, must touch)
- Problem Checklist (checkboxes, zero checked allowed):
  - Practical: Housing, Insurance, Transportation, Work/School
  - Family: Dealing with children, Dealing with partner
  - Emotional: Worry, Fears, Sadness, Depression, Nervousness, Loss of interest
  - Spiritual: Loss of faith, Loss of meaning
  - Physical: Pain, Nausea, Fatigue, Sleep problems, Eating, Bathing, Breathing, Mouth sores

? NCCN Distress Thermometer is a copyright-protected instrument. Confirm licensing before release.

6.4 Step 4 -- HADS (Hospital Anxiety & Depression Scale)

14 items total (A1-A7 Anxiety + D1-D7 Depression), each with 4 response options. Scored 0-3 per item.

- Anxiety subscale total: 0-21
- Depression subscale total: 0-21
- All 14 items must be answered before Submit.
- Option order matches the validated HADS instrument exactly (do not reorder).

? HADS is a copyright-protected instrument (GL Assessment). Confirm licensing before release.

Steps 1-4 are submitted together in a single API call.

6.5 Step 5 -- 10-Minute Resting HRV

- Triggered automatically after Step 1-4 submission.
- 10-minute countdown; patient must remain still with Polar H10 connected.
- Fields auto-filled from Polar during recording (not manual):
  - Heart Rate (bpm), RMSSD (ms), SDNN (ms), pNN50 (%), Mean HR (bpm)
- Submitted to backend separately after recording completes.
- Weekly assessment is fully_completed only after this step.
- Dashboard Weekly chip turns Done only after HRV submission.

6.6 Week 0 rules

- Week 0 form is mandatory before the first session starts.
- If Week 0 is not submitted, no session can be started.
- Week 0 contains the same 5 steps as other weekly forms.

6.7 Weekly form ordering rules

- Cannot submit Week N if Week N-1 was submitted in the current week (window still open).
- If Week N-1 window has expired (missed), Week N is unblocked.
- Institute cannot skip backwards (cannot submit Week 2 after Week 3 submitted).

7. Dashboard -- Action Button State Machine

The backend returns session/form state params per phase per patient. The frontend renders buttons accordingly:

Session status	Post Therapy form	Phase window	Button(s) shown
Not started	--	Open	Start
Not started	--	Closed	Missed (no action)
incomplete (user stopped)	Not submitted	Open	Restart + Resume
completed (auto)	Not submitted	Open	Resume
incomplete (user stopped)	Submitted	Open	Restart
completed (auto)	Submitted	Open	Done

Backend response params (per phase, per patient in list API):

```
{
  "phase": "phase1 | phase2 | phase3",
  "phaseStatus": "upcoming | active | missed",
  "sessionStatus": "not_started | incomplete | completed",
  "postFormStatus": "not_submitted | submitted",
  "actionButtons": ["start" | "restart" | "resume" | "done" | "missed"]
}
```

8. Dashboard -- Patient List

8.1 Sort order

- Default sort: Patients with pending sessions first (Phase 1 -> Phase 2 -> Phase 3), then completed, then missed last.
- Optional sorts: Compliance (low -> high), Enrollment Date.

8.2 Columns

Column	Source
Name & ID (SKITII ID)	Patient record (ID entered manually on enrollment)
Age	Patient record
Action (phase chips)	Backend phase state per patient
Group Type	Patient record (Adaptive / Control)
Weekly (Start / Done)	Weekly assessment status for current study week
Enrollment Date	Patient record
Treatment Duration	Patient record (3 Weeks / 6 Weeks)
Compliance Status	Computed: completedSessions / expectedSessions
Emergency Number	Patient record
Last Session	Last completed Post Therapy completedAt

8.3 Filters

- Date filter -- defaults to today; can select a specific date to view that day's session status.
- Phase chips: All | Phase 1 | Phase 2 | Phase 3 -- filters by session status for the selected phase on the selected date.
- Session status chips: All | Pending | Completed | Missed -- filters by session status for the selected date + phase.
- Weekly form chip: Weekly Pending | Weekly Submitted -- filters by current week's form status.
- Search -- by patient name only.
- Pagination -- server-side.

9. Add Patient (Enrollment)

9.1 Form sections

Personal Information

- First Name, Last Name
- Emergency Number* (India +91)
- Language Preference* (dropdown, e.g. Kannada)
- Age* (years, integer)
- Gender* (Male / Female / Other)

Clinical Data (Yenepoya v1 -- will differ per future hospital)

- Diagnosis Stage* (dropdown)
- Treatment Type* (dropdown, e.g. Both)
- Cancer Site* (text -- specific Head & Neck location)
- ECOG Performance Status* (dropdown)

Study Enrollment

- Enrollment Date* (date picker, dd/mm/yyyy)
- Expected Duration* (dropdown: 3 Weeks / 6 Weeks)
- Surgical Type* (dropdown, e.g. Post-Operative)
- Group Type* (dropdown: Adaptive / Control)

CTA: Enroll Patient (disabled until all fields valid)

9.2 Rules

- Patient is editable until Week 0 form is submitted.
- After Week 0 submission -> view only (no edits).
- Skitii ID (SKITII...) is the patient registration number, entered manually -- same as the Skitii product.
- Future hospitals: patient intake form will differ; backend must support per-institution schema versioning.

10. Statistics

Shown per institute (Yenepoya). Date range filter is mandatory -- options:

Option	Description
Today	Default
Last 3 Days	
Last 7 Days	
Last 2 Weeks	
Custom Date Range	User picks start and end date

4 KPI blocks displayed on the Statistics screen:

Block	Label	Definition
1	Registered Patients	Total enrolled patients
2	Sessions -- Today	Today's completed sessions, broken down by
3	Total Missed Sessions	Total sessions where phase window closed with
4	Total Missed Weekly Reports	Total study weeks elapsed with no fully_completed weekly assessment

Session = Post Therapy form submitted (not just session started). Missed session = phase window closed + no Post Therapy submitted for that phase.

11. Auth & PIN

11.1 Login flow

```
Login screen (email input)
    ? sendOtp (Existing Auth Backend)
OTP screen
    ? verifyOtp -> JWT
Create PIN (if not set on this device)
    OR
Enter PIN (if already set)
    ?
Dashboard
```

- PIN is 4 digits, stored in device Keychain/Keystore only (not on any server).
- PIN is per-device -- each tablet has its own PIN for the same institute account.
- No Administrator role in the trial app -- all logged-in institute users have equal access.

11.2 Forgot PIN

1. User taps Forgot PIN? on Enter PIN screen.
2. Trial Backend sends OTP to institute's login email (device-specific reset request).
3. User enters OTP on tablet.
4. On verify -> PIN cleared on that device only -> user is taken to Create PIN screen.
5. This resets the PIN for that device only. Other tablets are unaffected.

12. Control Music -- Admin Setup

1. In Skitii Admin Frontend, admin adds a flag controlPatientMusic: true to music assets.
2. These flagged tracks form the control music pool.
3. The GET /music API for control patients returns only flagged tracks.
4. On session start for a control patient: frontend picks one random track from the pool and plays it continuously.
5. If the patient changes music: another random track from the pool is selected and plays continuously.
6. Music ranking is not updated for control patients.
7. Session readings store the full music history per phase: [{ trackId, startedAt, endedAt }].

13. Trial Admin Panel

A new dedicated web admin panel built for managing the Skitii Trial. Separate from the existing Skitii Admin Frontend.

13.1 Responsibilities

Module	Description
Patient Management	View all enrolled patients, their details, group type (Adaptive / Control),
Session Data	View per-patient session history -- Pre Therapy, Baseline, Live Session, Post
Weekly Assessments	View submitted weekly form responses (Vitals, VAS, NCCN, HADS, HRV)
Statistics Dashboard	Admin-level overview of all KPIs across the trial (sessions, missed sessions,
Data Export	Export trial data (patients, sessions, assessments) as CSV or PDF for clinical
Institute Management	Add and manage hospital institutes and their trial assignments (supports
Trial Configuration	Configure trial parameters per institute (duration, patient intake schema,
	email reminders)

13.2 Access

- Accessible by Skitii internal team only (not the institute/hospital).
- Authenticated via existing Auth Backend (separate admin role).

15. Trial Backend -- API Summary

The Trial Backend serves both the React Native Mobile App (tablet) and the Trial Admin Panel. APIs listed here are for the mobile app; the Admin Panel will have additional endpoints for data management and exports.

API	Description
POST /auth/forgot-pin/send-otp	Send OTP to institute email for PIN reset (device-specific)
POST /auth/forgot-pin/verify-otp	Verify OTP, clear PIN for that device
POST /patients	Enroll new patient
GET /patients	List patients with phase state, weekly status, compliance (sorted)
GET /patients/:id	View patient detail
PATCH /patients/:id	Update patient (blocked after Week 0 submitted)
POST /sessions/start	Start session for a phase (validates phase window + ordering)
POST /sessions/:id/stop	Mark session as incomplete (user_stopped)
POST /sessions/:id/complete	Mark session as completed (auto_completed)
POST /sessions/:id/readings	Upload gzip readings chunk (Adaptive: per phase; Control: per music
POST /sessions/:id/post-therapy	Submit Post Therapy form
POST /weekly-assessments	Submit Steps 1-4 (batched)
POST /weekly-assessments/:id/hrv	Submit Step 5 HRV separately
GET /statistics	Fetch all KPI aggregates (filter: dateRange)
GET /music/control	Fetch control patient music pool

16. Key Technical Constraints

1. Tablet-only (landscape, Android/iOS -- OS TBD). No phone layout.
2. Polar H10 is the supported BLE device. Tablet reads RR intervals; HRV computed on-device.
3. On-device compute -- RMSSD, SDNN, pNN50 computed on tablet, uploaded to Trial Backend (same as product pattern).
4. Offline -- sessions require network for create/complete. Weekly forms prefer online. Offline queue TBD.
5. Concurrent multi-tablet -- Trial Backend must block a second session for the same patientId + phase + date if one is already live.
6. Phase 3 date attribution -- a session starting at 2AM July 18 is stored as Phase 3 of July 17 (sessionDate: 2027-07-17, phase: 3).
7. Future multi-hospital -- patient intake schema must be versioned per institutionId / trialId to support different fields for other hospitals.